

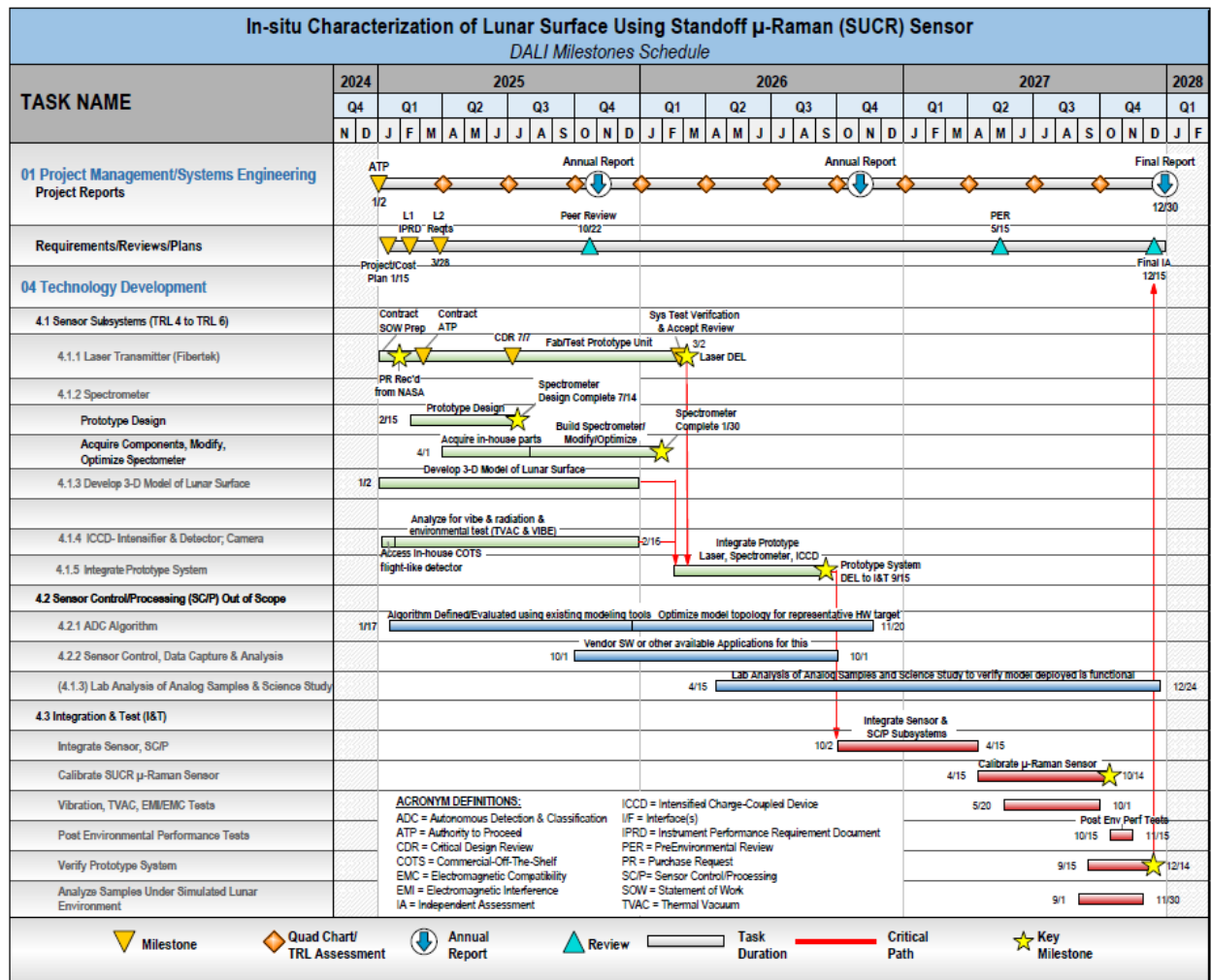
STATEMENT OF WORK FOR DEVELOPMENT OF SUCR

1.0 Introduction/Background: We have been awarded a research grant for the propose of developing to TRL-6 a Portable Pulsed Standoff Ultracompact μ -Raman (SUCR) sensor so that it can be proposed to future lunar missions. We propose to procure a space-qualifiable laser, an ICCD camera, and optical components, as well as fabricate spectrometer plates and to build a new sensor, by integrating those components and test its performance for use on a notional MMPACT mission. This proposed new sensor will be tested to Dusty TVAC system at NASA MSFC to verify performance.

This statement of work is the result of a proposal submitted by NASA Langley for an award under the NASA Headquarters Science Mission Directorate (SMD) Announcement of Opportunity entitled Development and Advancement of Lunar Instrumentation (DALI).

2.0 Scope of Work: This task will mature the SUCR sensor for robotic and compact lunar lander missions as described in the DALI call. DALI funds concept whose goals are applicable to human exploration, in situ resource utilization, and lunar science. SUCR is specifically designed to contribute to these goals and is flexible enough to be accommodated on multiple future lunar platforms. SUCR implementation supports NASA's broader lunar exploration goals, including human exploration and habitalization, ISRU, MMPACT DM-1 & DM-2 missions, CLPS, Artemis, and lunar science. The SUCR sensor is a prime candidate technology for missions focusing on lunar surface chemistry, mineralogy, and geology.

Dr. Beegle has developed compact remote laser-based systems for detecting lunar regolith (minerals/glasses), minerals, organics, and chemicals applications and applications suitable for NASA's landers and rovers via the SHERLOC instrument on the Perseverance rover. For the SUCR task, he will direct the TRL maturation, coordinate measurements of the breadboard and brassboard instrument and develop the requirements with Dr. Abedin.



3.0 Description of Tasks/Technical Requirements:

Task 3.1 Dr. Beegle will participate in monthly meetings, project status meetings and annual reviews as described in the following table:

MEETING	FREQUENCY/DATE	DURATION	LOCATION
Kick-Off Presentation	Within 1 month after contract award Date and time to be mutually agreed upon by the Contractor and COTR.	3 Days	At NASA Langley
Project Status Meetings	Weekly	1 Hour	Via Telecon
Annual Status Review	At the end of every work year, an annual status review will be conducted	2 Day	NASA Langley

Task 3.2 Dr. Beegle will participate in ad hoc meetings for the development of subsystems of the SUCR sensor to guide the development of these subsystems. He will communicate his experiences in the development of flight sensors to improve the TRL maturation process.

Task 3.3 Dr. Beegle will help design the test environmentally relevant experiments to validate the sensor in year 3 of the SUCR task. He will review the data, and make modifications to the experiments as necessary.

4.0 Deliverables:

In year 1, Dr. Beegle will develop 3-D modeling of the lunar surface for regolith or volatile species (e.g., water, ice, or hydroxyl) and produce a report defining the requirements of the SUCR flight sensor and deliver it to Dr. Abedin.

At the end of year 2 Dr. Beegle will produce a report commenting on the technological maturation of the SUCR sensor, including the identification of work to go, and help prioritize work to be done in year three.

During year 3, Dr. Beegle will help develop the test plan for the brassboard sensor that will be tested under lunar conditions. Additionally, he will help document all the performance metrics for the instrument.